

```
# =====
# Load Dataset
# =====

df = pd.read_csv("data.csv")

print("First 5 Rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nMissing Values:")
print(df.isnull().sum())
```

First 5 Rows:

	date	price	bedrooms	bathrooms	sqft_living	sqft_lot \
0	2014-05-02 00:00:00	313000.0	3.0	1.50	1340	7912
1	2014-05-02 00:00:00	2384000.0	5.0	2.50	3650	9050
2	2014-05-02 00:00:00	342000.0	3.0	2.00	1930	11947
3	2014-05-02 00:00:00	420000.0	3.0	2.25	2000	8030
4	2014-05-02 00:00:00	550000.0	4.0	2.50	1940	10500

	floors	waterfront	view	condition	sqft_above	sqft_basement	yr_built \
0	1.5	0	0	3	1340	0	1955
1	2.0	0	4	5	3370	280	1921
2	1.0	0	0	4	1930	0	1966
3	1.0	0	0	4	1000	1000	1963
4	1.0	0	0	4	1140	800	1976

	yr_renovated	street	city	statezip	country
0	2005	18810 Densmore Ave N	Shoreline	WA 98133	USA
1	0	709 W Blaine St	Seattle	WA 98119	USA
2	0	26206-26214 143rd Ave SE	Kent	WA 98042	USA
3	0	857 170th Pl NE	Bellevue	WA 98008	USA
4	1992	9105 170th Ave NE	Redmond	WA 98052	USA

Dataset Shape:
(4600, 18)

```
Missing Values:
date           0
price          0
bedrooms       0
bathrooms      0
sqft_living    0
sqft_lot       0
floors         0
waterfront     0
view           0
condition      0
sqft_above     0
sqft_basement  0
yr_built       0
yr_renovated   0
street         0
city           0
statezip       0
country        0
dtype: int64
```

```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("\n===== MODEL PERFORMANCE =====")

print("Mean Absolute Error (MAE):", mae)
print("Mean Squared Error (MSE):", mse)
print("Root Mean Squared Error (RMSE):", rmse)
print("R2 Score:", r2)
```

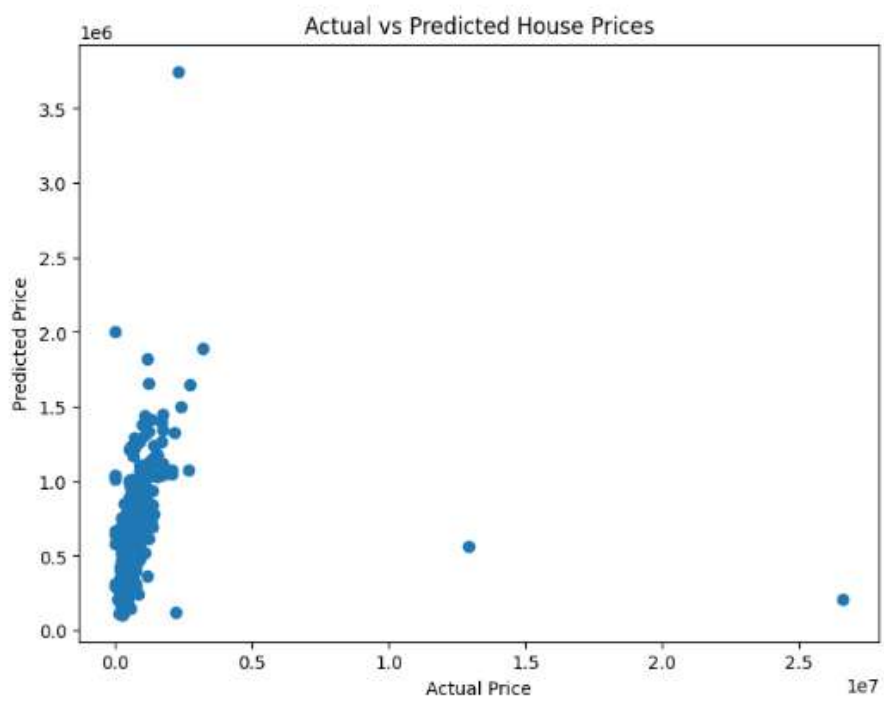
```
===== MODEL PERFORMANCE =====
Mean Absolute Error (MAE): 207555.78906405612
Mean Squared Error (MSE): 985909494893.7739
Root Mean Squared Error (RMSE): 992929.7532523506
R2 Score: 0.03327642984950774
```

===== COEFFICIENTS =====

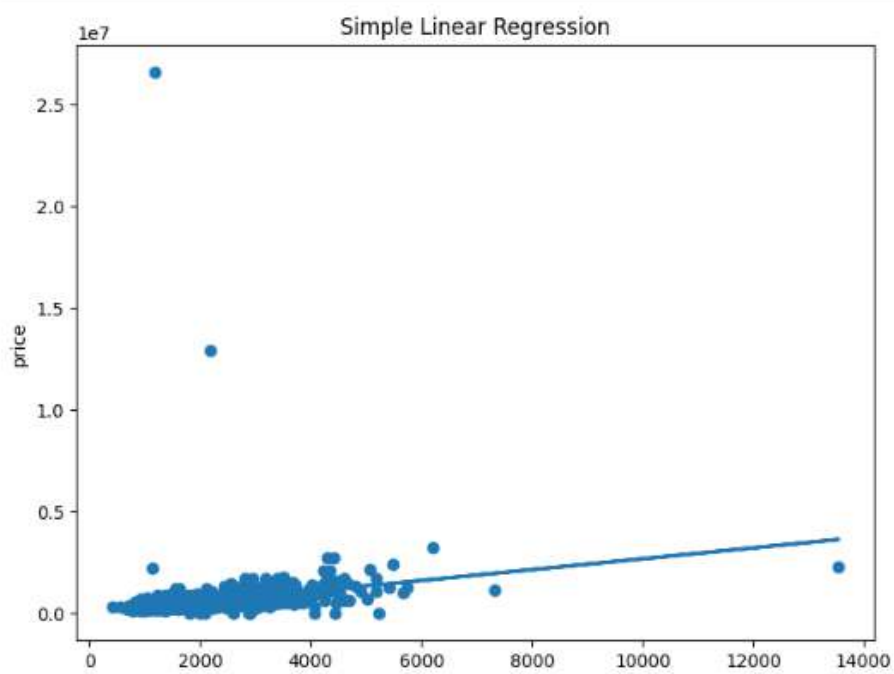
	Feature	Coefficient
5	waterfront	381090.451525
4	floors	63769.947303
6	view	48833.598053
1	bathrooms	35484.169820
7	condition	29798.074791
12	city	3062.436963
2	sqft_living	184.614531
8	sqft_above	95.700277
9	sqft_basement	88.914254
11	yr_renovated	9.102008
3	sqft_lot	-0.537507
13	statezip	-1404.116524
10	yr_built	-2549.026519
0	bedrooms	-62089.383207

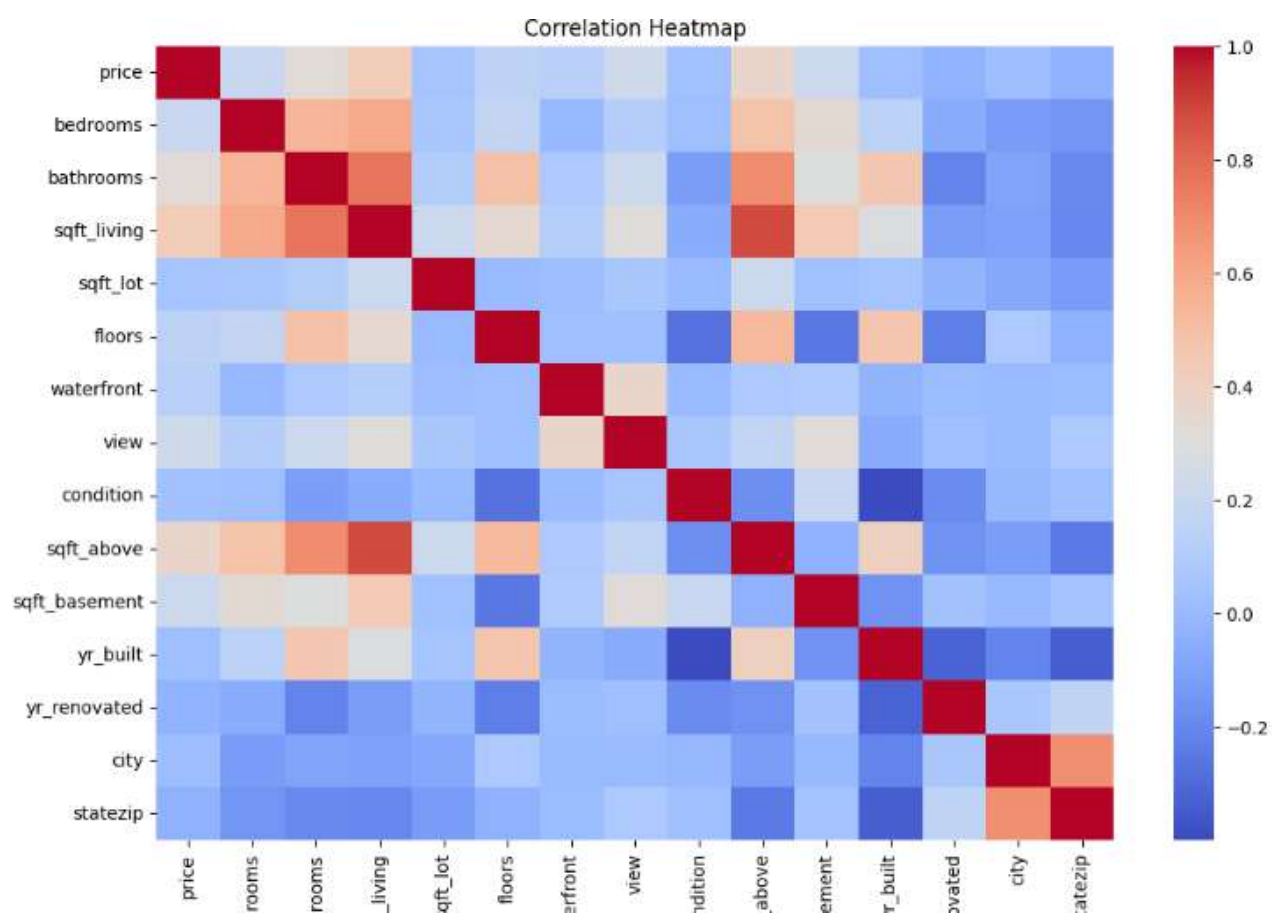
Intercept:

4867135.15302121



price: show /





```
===== SAMPLE PREDICTIONS =====  
   Actual Price Predicted Price  
0  5.440000e+05  3.056626e+05  
1  0.000000e+00  3.117828e+05  
2  1.712500e+06  1.081407e+06  
3  3.650000e+05  5.707895e+05  
4  2.750000e+05  3.834782e+05  
5  6.250000e+05  5.928231e+05  
6  4.530000e+05  4.603857e+05  
7  3.000000e+05  4.200372e+05  
8  4.179857e+05  5.094399e+05  
9  6.725000e+05  5.386652e+05
```

In []: